AQA 3.2.4 Properties of Period 3 elements and their oxides

AQA A-level Chemistry · Revision questions · Sheet 3 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- Melting points: giant ionic and giant covalent oxides are high; simple molecular oxides are low (weak van der Waals forces)
- $Mg + H_2O(g) \rightarrow MgO + H_2$ (steam); $2Na + 2H_2O \rightarrow 2NaOH + H_2$
- Basic oxides react with acids to give salt + water; acidic oxides react with bases (e.g. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$)
- P_4O_{10}, SO_2, SO_3 are simple molecular and acidic: $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 \rightarrow H_2SO_3$; $SO_3 \rightarrow H_2SO_4$
- Na_2O and MgO are giant ionic and basic: $Na_2O + H_2O \rightarrow 2NaOH$ (pH 12–14); $MgO + H_2O \rightarrow Mg(OH)_2$ (pH 9–10)

Section A: Short-answer questions

01 0.10 mol of each oxide is added to separate 1.0 dm^3 samples of water. Which forms the solution with the highest pH?

[1 mark]

A Na_2O
B MgO
C SiO_2
D P_4O_{10}

02 Which element forms an ionic oxide that reacts with strong alkalis?

[1 mark]

A Aluminium
B Magnesium
C Sodium
D Sulfur

03 Suggest why phosphorus(V) oxide, P_4O_{10}, has a higher melting point than sulfur trioxide, SO_3.

[2 marks]

04 Aluminium oxide has a lower melting point than would be expected for a purely ionic compound. Suggest why.

[2 marks]

05 Give an equation for the complete neutralisation of phosphoric acid, H_3PO_4, by sodium hydroxide.

[1 mark]

Section B: Longer questions

06 Silicon dioxide and sulfur trioxide are oxides of Period 3 elements.

06.1 State the type of crystal structure shown in silicon dioxide and in sulfur trioxide.

[2 marks]

06.2 Explain why silicon dioxide has a higher melting point than sulfur trioxide.

[4 marks]

07 This question is about reactions of Period 3 elements with water and oxygen.

07.1 Give an equation for the reaction of sodium with cold water. State **one** observation.

[2 marks]

07.2 Magnesium reacts very slowly with cold water. Give an equation for this reaction and suggest the pH of the solution formed.

[2 marks]

07.3 Give equations for the reactions of phosphorus and of sulfur with oxygen.

[2 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**A**	*Source: Practice question (written for this set)*	1
02	**A**	*Source: 7405/3 Jun 2018 Q25*	1
03	M1 P_4O_{10} molecules are larger / have more electrons M2 so stronger van der Waals forces between molecules	*Source: Practice question (written for this set)*	2
04	M1 Al^{3+} is small with a high charge density / is very polarising M2 so the oxide ion is distorted and the bonding has some covalent character	*Source: Practice question (written for this set)*	2
05	$H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$	*Source: Practice question (written for this set)*	1
06.1	SiO_2 giant covalent / macromolecular SO_3 molecular / (simple) molecule	*Source: 7405/1 Jun 2017 Q06.3–06.4* Do not allow simple covalent	2
06.2	M1 covalent bonds (between atoms) in SiO_2 M2 van der Waals / intermolecular forces between molecules in SO_3 M3 covalent bonds are stronger than van der Waals forces M4 (covalent bonds) take more energy to overcome / break	If metallic or ionic, max 1 If IMF in SiO_2, max 1 (M2 only) Allow dipole–dipole forces between molecules	4
07.1	$2Na + 2H_2O \rightarrow 2NaOH + H_2$ fizzes / moves around on the surface / melts into a ball / dissolves	*Source: Practice question (written for this set)*	2
07.2	$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$ pH 9–10		2
07.3	$P_4 + 5O_2 \rightarrow P_4O_{10}$ (allow $4P + 5O_2$) $S + O_2 \rightarrow SO_2$		2